

Assignment 4 :

GitHub Link :

https://github.com/ssbagi/IBM_OpenPOWER_and_SoC_Design/tree/main/DAY2/Assignment%204

README.md File :

https://github.com/ssbagi/IBM_OpenPOWER_and_SoC_Design/blob/main/DAY2/Assignment%204/README.md

Question : Design and simulate a 4-bit Up/Down Counter using Verilog HDL.

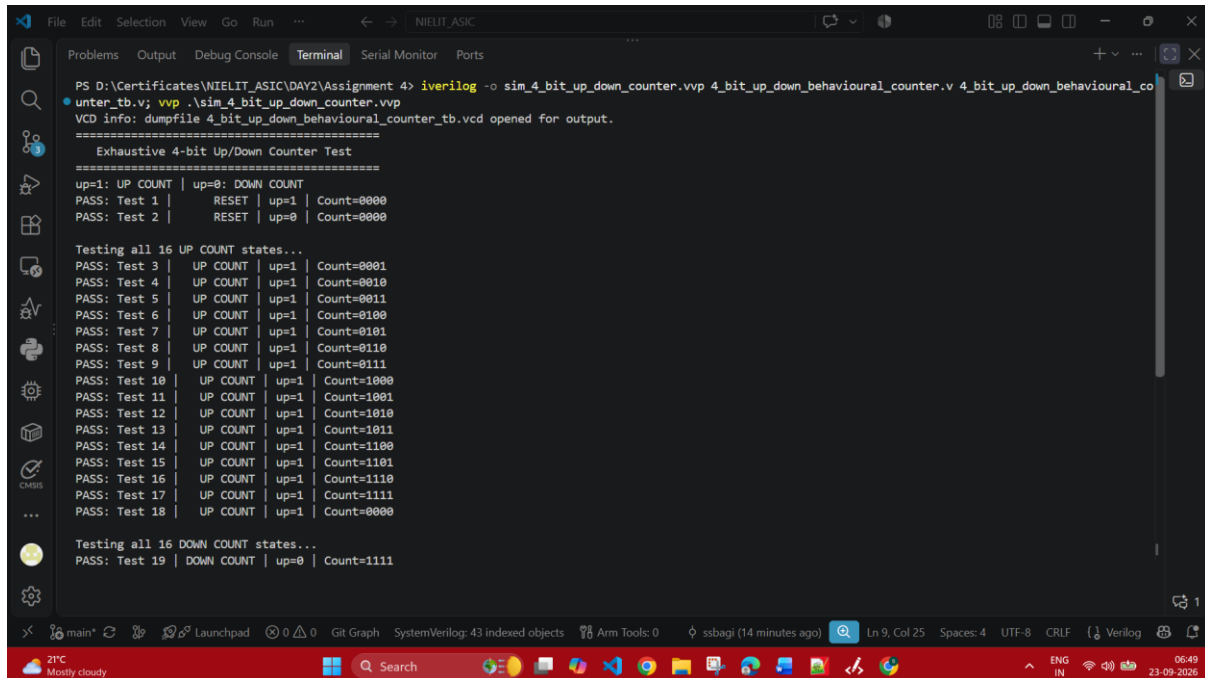
- a) Behavioural modeling.
- b) Synchronous reset implementation.
- c) Asynchronous reset implementation.
- d) Verify up-counting operation.
- e) Verify down-counting operation.

a) Behavioural modeling :

```
module up_down_counter_4_bit (  
    input    clk,  
    input    reset,  
    input    up,  
    output reg [3:0] count  
);  
    always @(posedge clk) begin  
        if (reset)  
            count <= 4'b0000;  
        else if (up)  
            count <= count + 4'b0001;  
        else  
            count <= count - 4'b0001;  
    end
```

endmodule

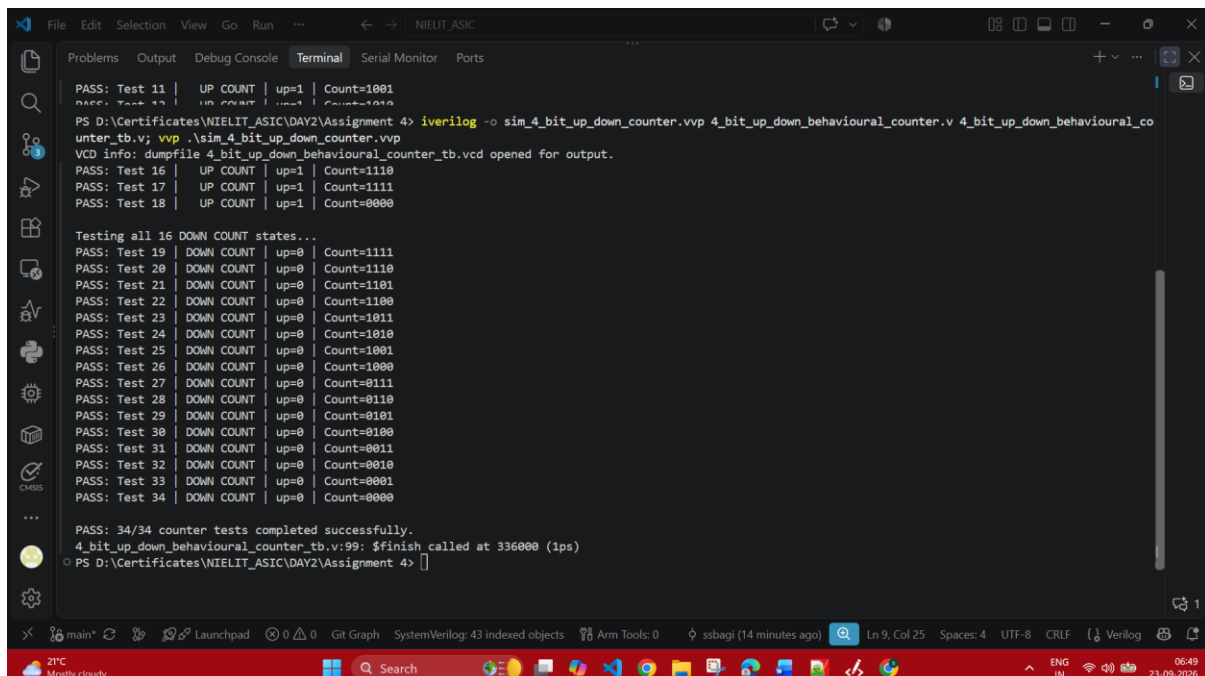
Compilation :



```
PS D:\Certificates\NIELIT_ASIC\DAY2\Assignment 4> iverilog -o sim_4_bit_up_down_counter.vvp 4_bit_up_down_behavioural_counter.v 4_bit_up_down_behavioural_counter_tb.v; vvp .\sim_4_bit_up_down_counter.vvp
VCD info: dumpfile 4_bit_up_down_behavioural_counter_tb.vcd opened for output.
=====
Exhaustive 4-bit Up/Down Counter Test
=====
up=1: UP COUNT | up=0: DOWN COUNT
PASS: Test 1 | RESET | up=1 | Count=0000
PASS: Test 2 | RESET | up=0 | Count=0000

Testing all 16 UP COUNT states...
PASS: Test 3 | UP COUNT | up=1 | Count=0001
PASS: Test 4 | UP COUNT | up=1 | Count=0010
PASS: Test 5 | UP COUNT | up=1 | Count=0011
PASS: Test 6 | UP COUNT | up=1 | Count=0100
PASS: Test 7 | UP COUNT | up=1 | Count=0101
PASS: Test 8 | UP COUNT | up=1 | Count=0110
PASS: Test 9 | UP COUNT | up=1 | Count=0111
PASS: Test 10 | UP COUNT | up=1 | Count=1000
PASS: Test 11 | UP COUNT | up=1 | Count=1001
PASS: Test 12 | UP COUNT | up=1 | Count=1010
PASS: Test 13 | UP COUNT | up=1 | Count=1011
PASS: Test 14 | UP COUNT | up=1 | Count=1100
PASS: Test 15 | UP COUNT | up=1 | Count=1101
PASS: Test 16 | UP COUNT | up=1 | Count=1110
PASS: Test 17 | UP COUNT | up=1 | Count=1111
PASS: Test 18 | UP COUNT | up=1 | Count=0000

Testing all 16 DOWN COUNT states...
PASS: Test 19 | DOWN COUNT | up=0 | Count=1111
```



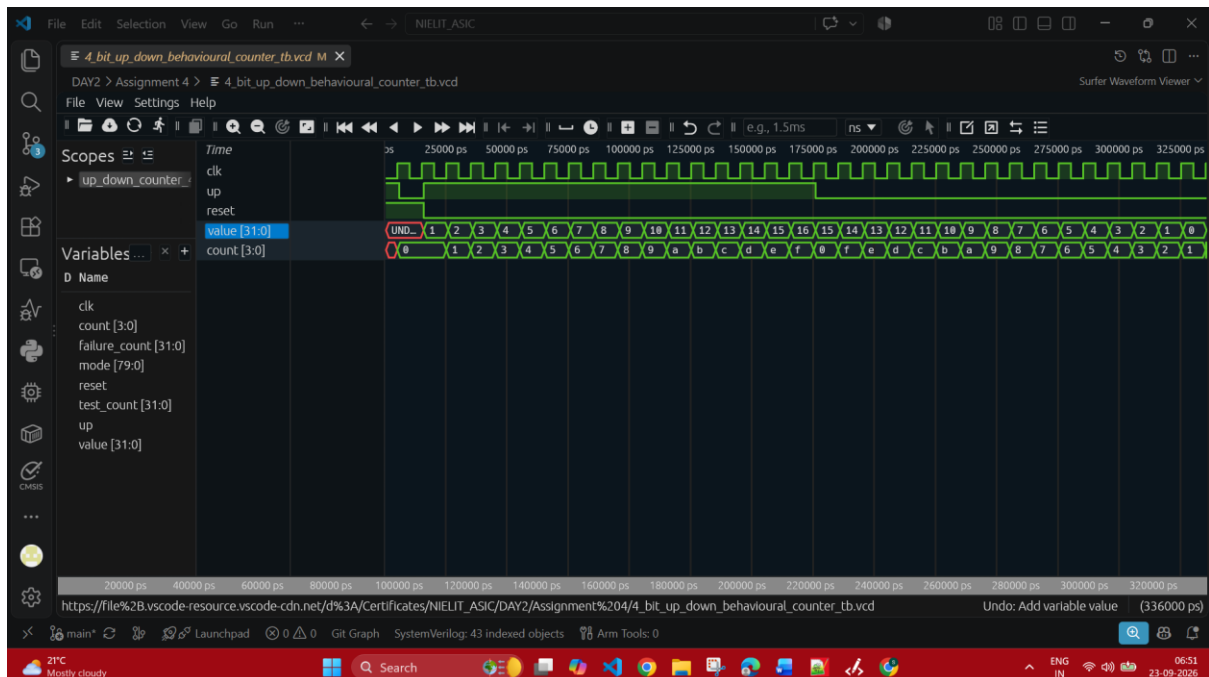
```
PASS: Test 11 | UP COUNT | up=1 | Count=1001
PASS: Test 12 | UP COUNT | up=1 | Count=1010
PASS: Test 13 | UP COUNT | up=1 | Count=1011
PASS: Test 14 | UP COUNT | up=1 | Count=1100
PASS: Test 15 | UP COUNT | up=1 | Count=1101
PASS: Test 16 | UP COUNT | up=1 | Count=1110
PASS: Test 17 | UP COUNT | up=1 | Count=1111
PASS: Test 18 | UP COUNT | up=1 | Count=0000

Testing all 16 DOWN COUNT states...
PASS: Test 19 | DOWN COUNT | up=0 | Count=1111
PASS: Test 20 | DOWN COUNT | up=0 | Count=1110
PASS: Test 21 | DOWN COUNT | up=0 | Count=1101
PASS: Test 22 | DOWN COUNT | up=0 | Count=1100
PASS: Test 23 | DOWN COUNT | up=0 | Count=1011
PASS: Test 24 | DOWN COUNT | up=0 | Count=1010
PASS: Test 25 | DOWN COUNT | up=0 | Count=1001
PASS: Test 26 | DOWN COUNT | up=0 | Count=1000
PASS: Test 27 | DOWN COUNT | up=0 | Count=0111
PASS: Test 28 | DOWN COUNT | up=0 | Count=0110
PASS: Test 29 | DOWN COUNT | up=0 | Count=0101
PASS: Test 30 | DOWN COUNT | up=0 | Count=0100
PASS: Test 31 | DOWN COUNT | up=0 | Count=0011
PASS: Test 32 | DOWN COUNT | up=0 | Count=0010
PASS: Test 33 | DOWN COUNT | up=0 | Count=0001
PASS: Test 34 | DOWN COUNT | up=0 | Count=0000

PASS: 34/34 counter tests completed successfully.
4_bit_up_down_behavioural_counter_tb.v:99: $finish called at 336000 (1ps)
PS D:\Certificates\NIELIT_ASIC\DAY2\Assignment 4>
```

Simulation :

Synchronous reset. In the next clk edge only the count increment is done and stuff.



b) Synchronous reset implementation :

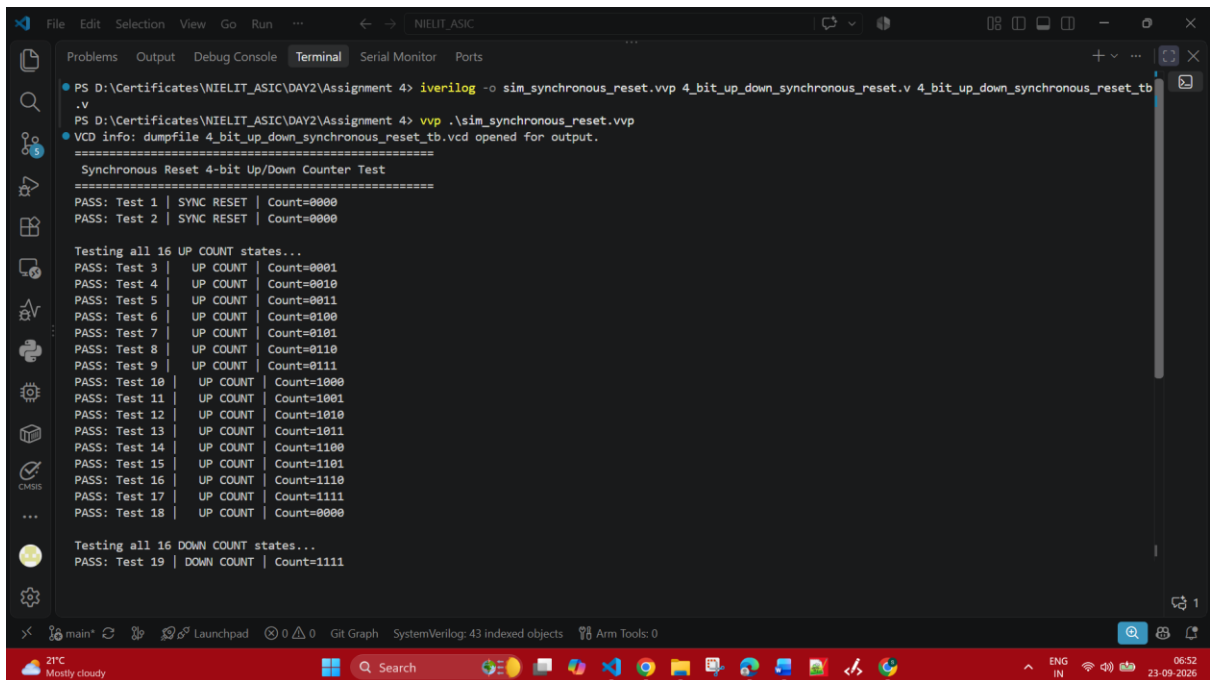
Verilog code :

```
module up_down_counter_4_bit (
    input      clk,
    input      reset,
    input      up,
    output reg [3:0] count
);

always @(posedge clk) begin
    if (reset)
        count <= 4'b0000;
    else if (up)
        count <= count + 4'b0001;
    else
        count <= count - 4'b0001;
    end

endmodule
```

Compilation :

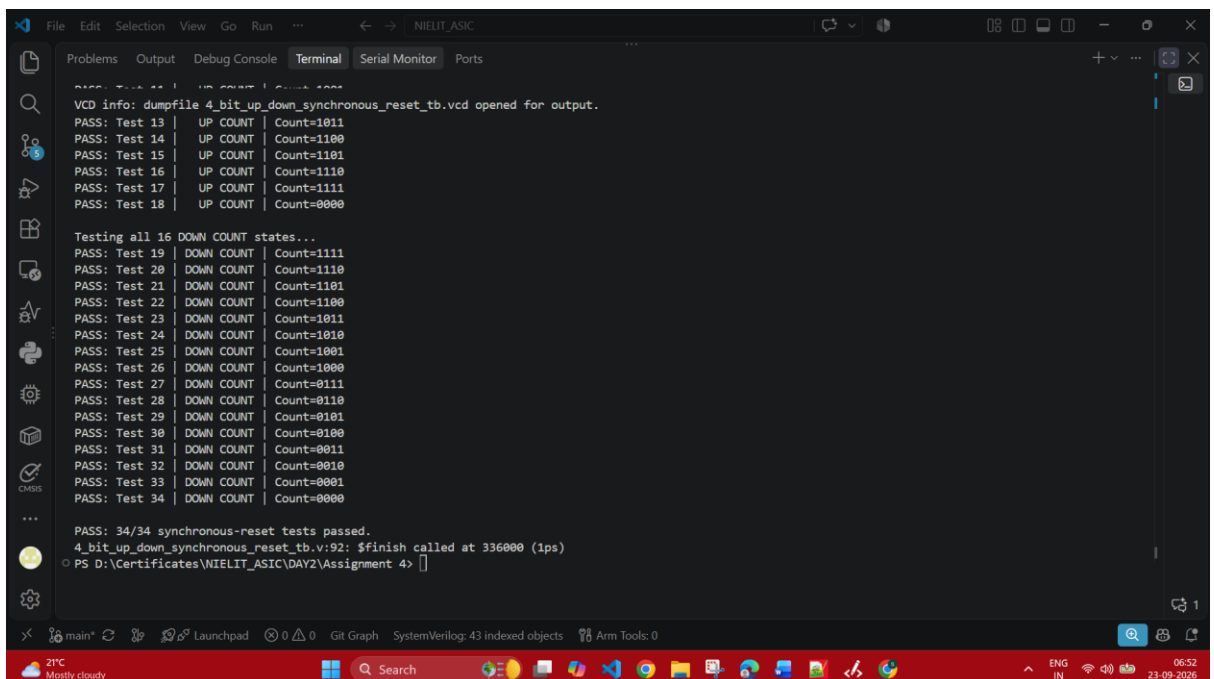


The screenshot shows a terminal window with the following output:

```
PS D:\Certificates\NIELIT_ASIC\DAY2\Assignment 4> iverilog -o sim_synchronous_reset.vvp 4_bit_up_down_synchronous_reset.v 4_bit_up_down_synchronous_reset_tb.v
PS D:\Certificates\NIELIT_ASIC\DAY2\Assignment 4> vvp .\sim_synchronous_reset.vvp
VCD info: dumpfile 4_bit_up_down_synchronous_reset_tb.vcd opened for output.
=====
Synchronous Reset 4-bit Up/Down Counter Test
=====
PASS: Test 1 | SYNC RESET | Count=0000
PASS: Test 2 | SYNC RESET | Count=0000

Testing all 16 UP COUNT states...
PASS: Test 3 | UP COUNT | Count=0001
PASS: Test 4 | UP COUNT | Count=0010
PASS: Test 5 | UP COUNT | Count=0011
PASS: Test 6 | UP COUNT | Count=0100
PASS: Test 7 | UP COUNT | Count=0101
PASS: Test 8 | UP COUNT | Count=0110
PASS: Test 9 | UP COUNT | Count=0111
PASS: Test 10 | UP COUNT | Count=1000
PASS: Test 11 | UP COUNT | Count=1001
PASS: Test 12 | UP COUNT | Count=1010
PASS: Test 13 | UP COUNT | Count=1011
PASS: Test 14 | UP COUNT | Count=1100
PASS: Test 15 | UP COUNT | Count=1101
PASS: Test 16 | UP COUNT | Count=1110
PASS: Test 17 | UP COUNT | Count=1111
PASS: Test 18 | UP COUNT | Count=0000

Testing all 16 DOWN COUNT states...
PASS: Test 19 | DOWN COUNT | Count=1111
```



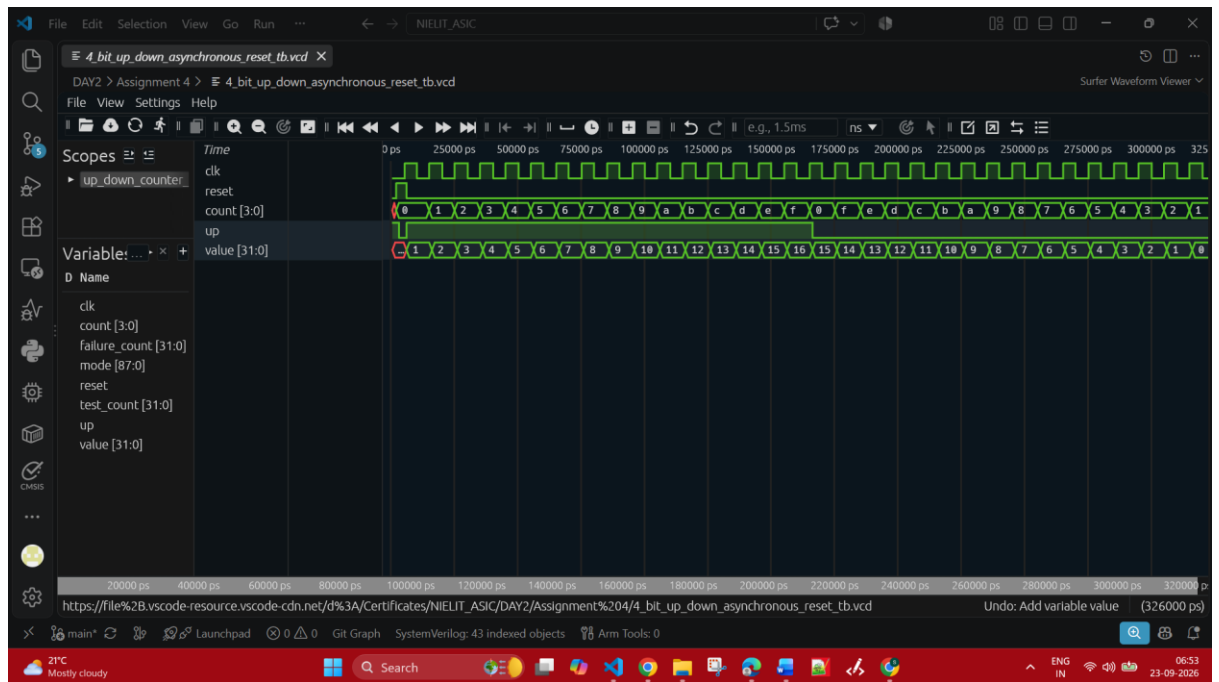
The screenshot shows the continuation of the simulation process in a terminal window. The output includes:

```
VCD info: dumpfile 4_bit_up_down_synchronous_reset_tb.vcd opened for output.
PASS: Test 13 | UP COUNT | Count=1011
PASS: Test 14 | UP COUNT | Count=1100
PASS: Test 15 | UP COUNT | Count=1101
PASS: Test 16 | UP COUNT | Count=1110
PASS: Test 17 | UP COUNT | Count=1111
PASS: Test 18 | UP COUNT | Count=0000

Testing all 16 DOWN COUNT states...
PASS: Test 19 | DOWN COUNT | Count=1111
PASS: Test 20 | DOWN COUNT | Count=1110
PASS: Test 21 | DOWN COUNT | Count=1101
PASS: Test 22 | DOWN COUNT | Count=1100
PASS: Test 23 | DOWN COUNT | Count=1011
PASS: Test 24 | DOWN COUNT | Count=1010
PASS: Test 25 | DOWN COUNT | Count=1001
PASS: Test 26 | DOWN COUNT | Count=1000
PASS: Test 27 | DOWN COUNT | Count=0111
PASS: Test 28 | DOWN COUNT | Count=0110
PASS: Test 29 | DOWN COUNT | Count=0101
PASS: Test 30 | DOWN COUNT | Count=0100
PASS: Test 31 | DOWN COUNT | Count=0011
PASS: Test 32 | DOWN COUNT | Count=0010
PASS: Test 33 | DOWN COUNT | Count=0001
PASS: Test 34 | DOWN COUNT | Count=0000

PASS: 34/34 synchronous-reset tests passed.
4_bit_up_down_synchronous_reset_tb.v:92: $finish called at 336000 (1ps)
PS D:\Certificates\NIELIT_ASIC\DAY2\Assignment 4>
```

Simulation :

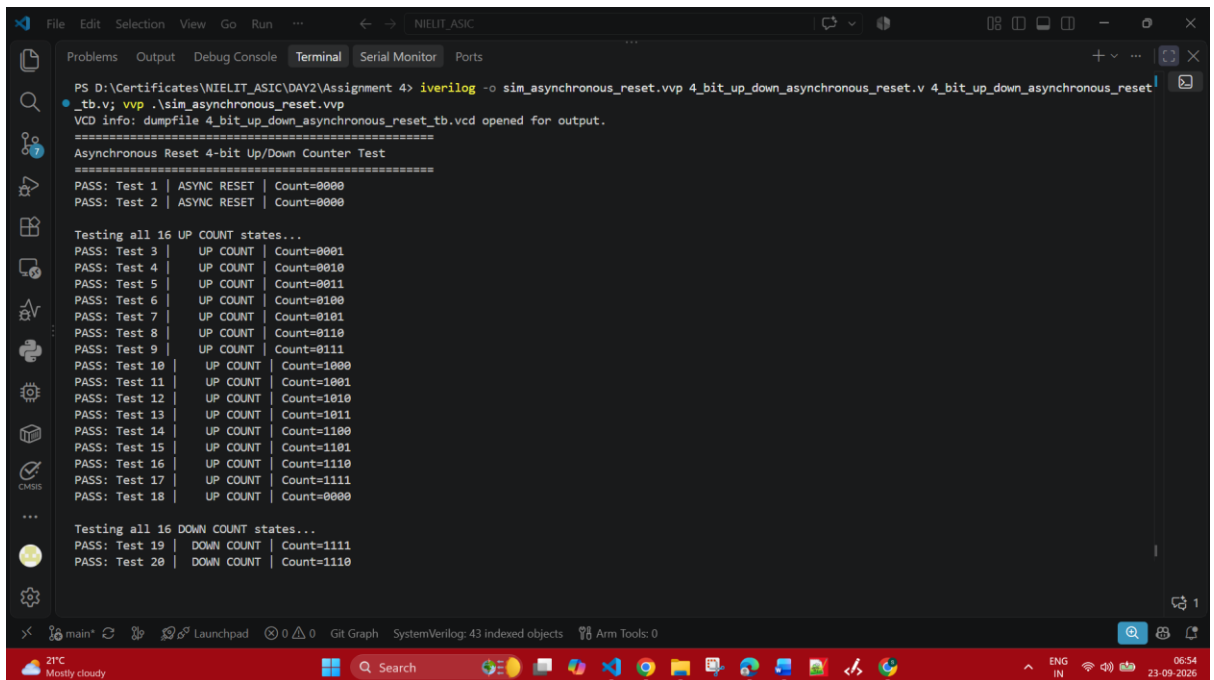


c) Asynchronous reset implementation :

Verilog code :

```
module up_down_counter_asynchronous_reset (
    input      clk,
    input      reset,
    input      up,
    output reg [3:0] count
);
    always @(posedge clk or posedge reset) begin
        if (reset)
            count <= 4'b0000;
        else if (up)
            count <= count + 4'b0001;
        else
            count <= count - 4'b0001;
        end
    endmodule
```

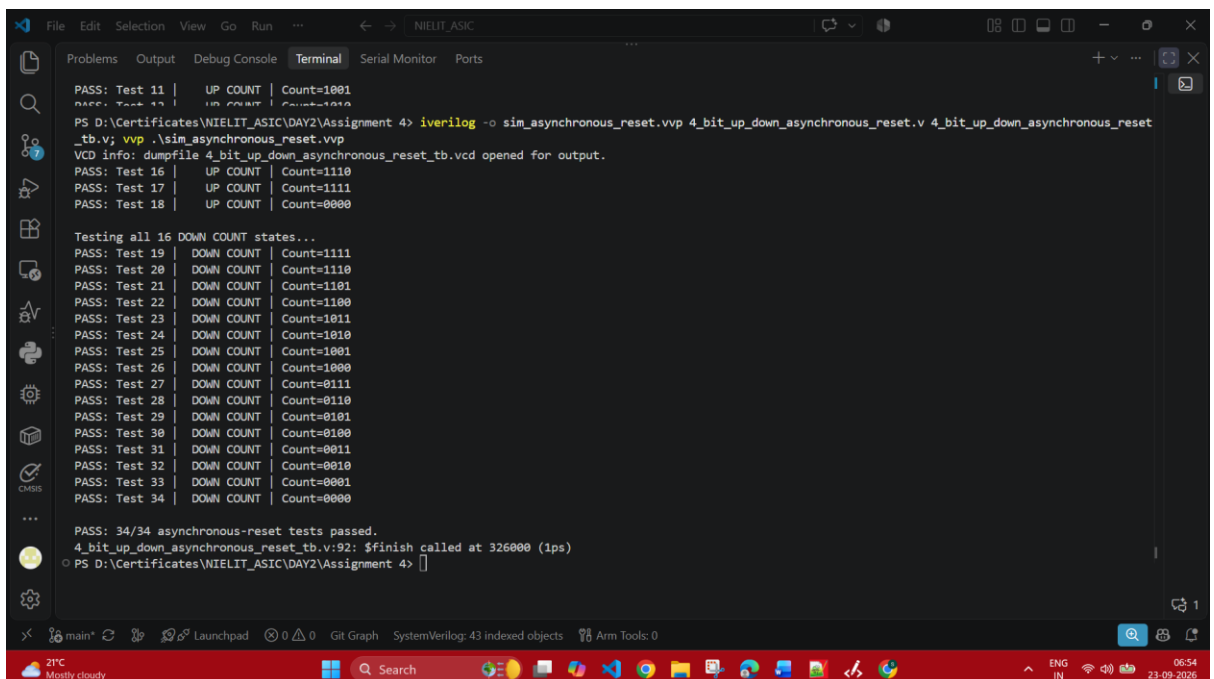
Compilation :



```
PS D:\Certificates\NIELIT_ASIC\DAY2\Assignment 4> iverilog -o sim_asynchronous_reset.vvp 4_bit_up_down_asynchronous_reset.v 4_bit_up_down_asynchronous_reset_tb.v; vvp .\sim_asynchronous_reset.vvp
VCD info: dumpfile 4_bit_up_down_asynchronous_reset_tb.vcd opened for output.
=====
Asynchronous Reset 4-bit Up/Down Counter Test
=====
PASS: Test 1 | ASYNC RESET | Count=0000
PASS: Test 2 | ASYNC RESET | Count=0000

Testing all 16 UP COUNT states...
PASS: Test 3 | UP COUNT | Count=0001
PASS: Test 4 | UP COUNT | Count=0010
PASS: Test 5 | UP COUNT | Count=0011
PASS: Test 6 | UP COUNT | Count=0100
PASS: Test 7 | UP COUNT | Count=0101
PASS: Test 8 | UP COUNT | Count=0110
PASS: Test 9 | UP COUNT | Count=0111
PASS: Test 10 | UP COUNT | Count=1000
PASS: Test 11 | UP COUNT | Count=1001
PASS: Test 12 | UP COUNT | Count=1010
PASS: Test 13 | UP COUNT | Count=1011
PASS: Test 14 | UP COUNT | Count=1100
PASS: Test 15 | UP COUNT | Count=1101
PASS: Test 16 | UP COUNT | Count=1110
PASS: Test 17 | UP COUNT | Count=1111
PASS: Test 18 | UP COUNT | Count=0000

Testing all 16 DOWN COUNT states...
PASS: Test 19 | DOWN COUNT | Count=1111
PASS: Test 20 | DOWN COUNT | Count=1110
```



```
PASS: Test 11 | UP COUNT | Count=1001
PASS: Test 12 | UP COUNT | Count=1010
PASS: Test 13 | UP COUNT | Count=1011
PASS: Test 14 | UP COUNT | Count=1100
PASS: Test 15 | UP COUNT | Count=1101
PASS: Test 16 | UP COUNT | Count=1110
PASS: Test 17 | UP COUNT | Count=1111
PASS: Test 18 | UP COUNT | Count=0000

Testing all 16 DOWN COUNT states...
PASS: Test 19 | DOWN COUNT | Count=1111
PASS: Test 20 | DOWN COUNT | Count=1110
PASS: Test 21 | DOWN COUNT | Count=1101
PASS: Test 22 | DOWN COUNT | Count=1100
PASS: Test 23 | DOWN COUNT | Count=1011
PASS: Test 24 | DOWN COUNT | Count=1010
PASS: Test 25 | DOWN COUNT | Count=1001
PASS: Test 26 | DOWN COUNT | Count=1000
PASS: Test 27 | DOWN COUNT | Count=0111
PASS: Test 28 | DOWN COUNT | Count=0110
PASS: Test 29 | DOWN COUNT | Count=0101
PASS: Test 30 | DOWN COUNT | Count=0100
PASS: Test 31 | DOWN COUNT | Count=0011
PASS: Test 32 | DOWN COUNT | Count=0010
PASS: Test 33 | DOWN COUNT | Count=0001
PASS: Test 34 | DOWN COUNT | Count=0000

...
PASS: 34/34 asynchronous-reset tests passed.
4_bit_up_down_asynchronous_reset_tb.v:92: $finish called at 326000 (1ps)
PS D:\Certificates\NIELIT_ASIC\DAY2\Assignment 4>
```

Simulation :

